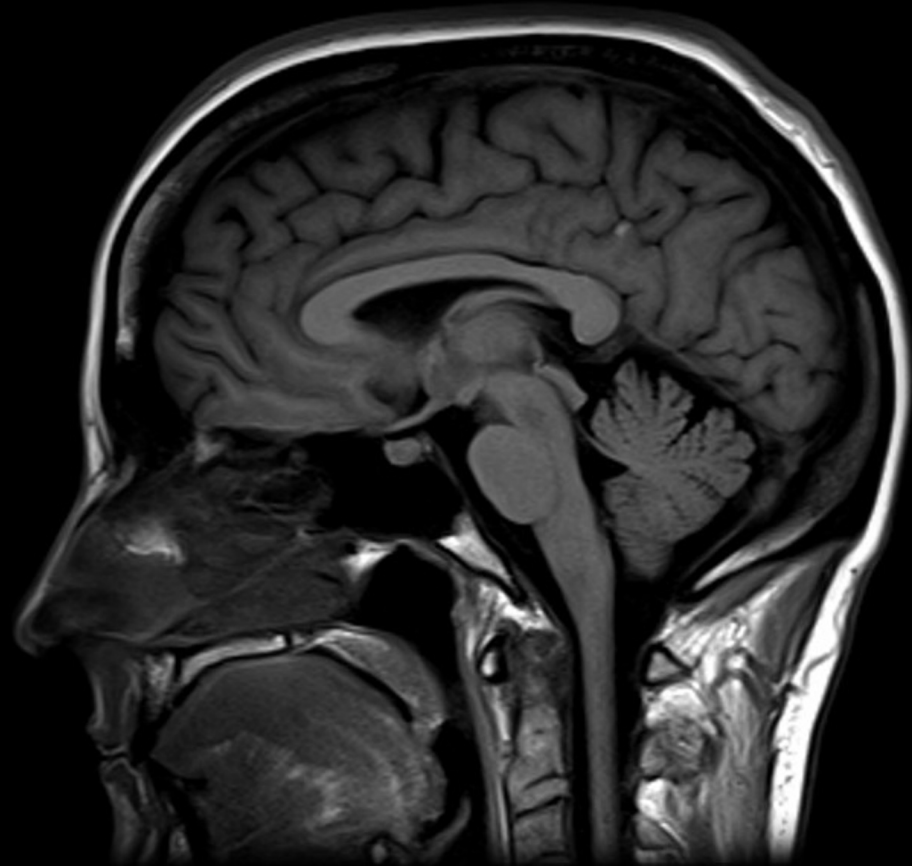


# Brain tumor classification

Variational Quantum Classifier  
with Qiskit

Dr. Boubacar Tanda Bonkano



# Speaker



**Dr. rer. nat. Boubacar Tanda Bonkano**

Fritz Haber Institute of the Max-Planck Society  
and TU Berlin (Alumni)

Experimental Physicist

Hybrid Semiconductor Systems

Electron dynamics

Ultrafast Time-Resolved Optical Spectroscopy

Time-Correlated Single Photon Counting

Data Science & Machine Learning

## **Scientific Interests:**

Photonics, Sensing , and Quantum Technologies

# Context and Motivation

## **What are brain tumors?**

Brain tumors are an abnormal aggregation of cells that arise in the central nervous system.

Brain tumors: non-cancerogenic or cancerogenic.

**Survival of patients depends on:** Early detection and fast diagnosis!

**Challenge:** Manual Brain tumor labeling is slow!

**Approach:** Development of classification and segmentation models to accelerate the diagnosis

# Context and Motivation

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**Challenge:** Manual Brain tumor labeling is slow!

**Approach:** Development of classification and segmentation models to accelerate the diagnosis



**Data-driven solution : Model based automatic detection**

# Data Source

Data source: Kaggle (Brain tumor MRI dataset)

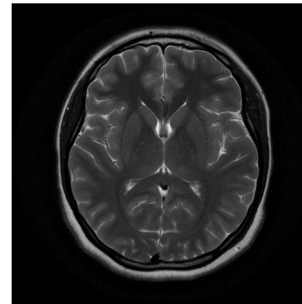
Original Data type: MRI images (.jpg)

MRI images: Without tumor, with tumor

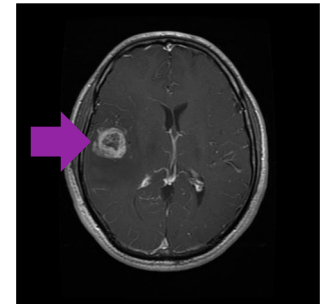
MRI image tumor types:

- Glioma
- Meningioma
- Pituitary

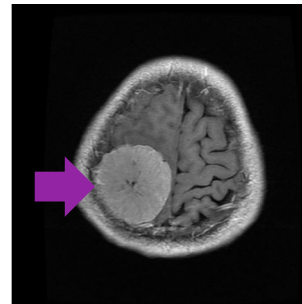
**No tumor**



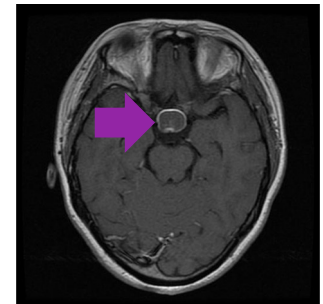
**Glioma**



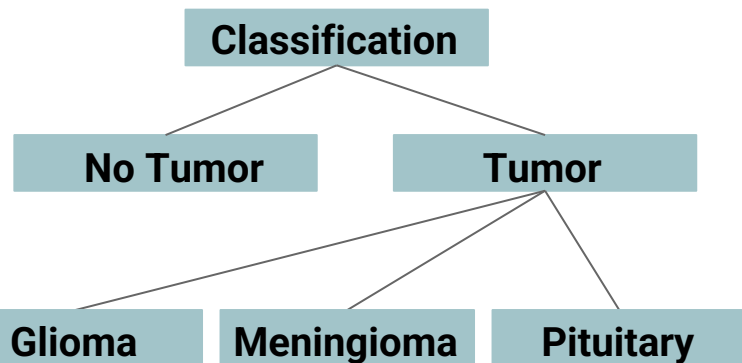
**Meningioma**



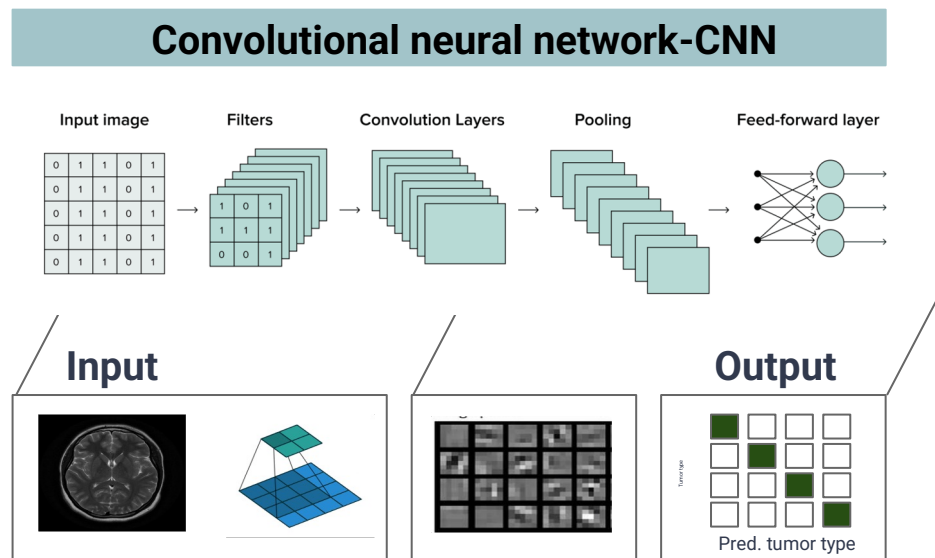
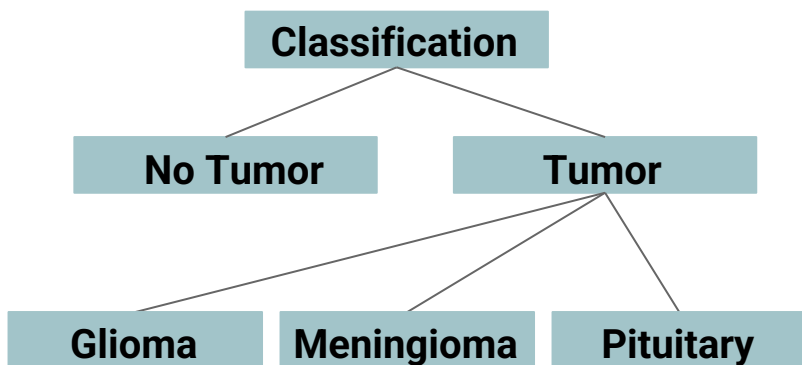
**Pituitary**



# Classical Approach



# Classical Approach



# Hybrid CNN-Quantum Approach

**Input**

Image

Features Extraction

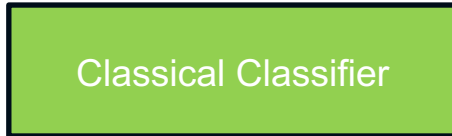
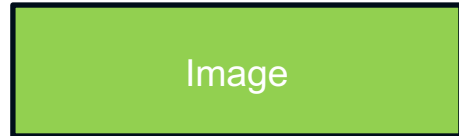
Classical Classifier

**Classical Output**

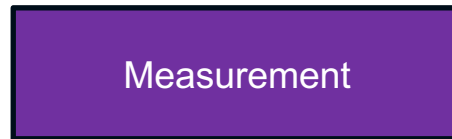
Prediction

# Hybrid CNN-Quantum Approach

**Input**



**Classical Output**



**Quantum Output**

# Hybrid CNN-Quantum Approach

## **Application Session:**

- A. QSVC**
- B. VQC**

# Hybrid CNN-Quantum Approach

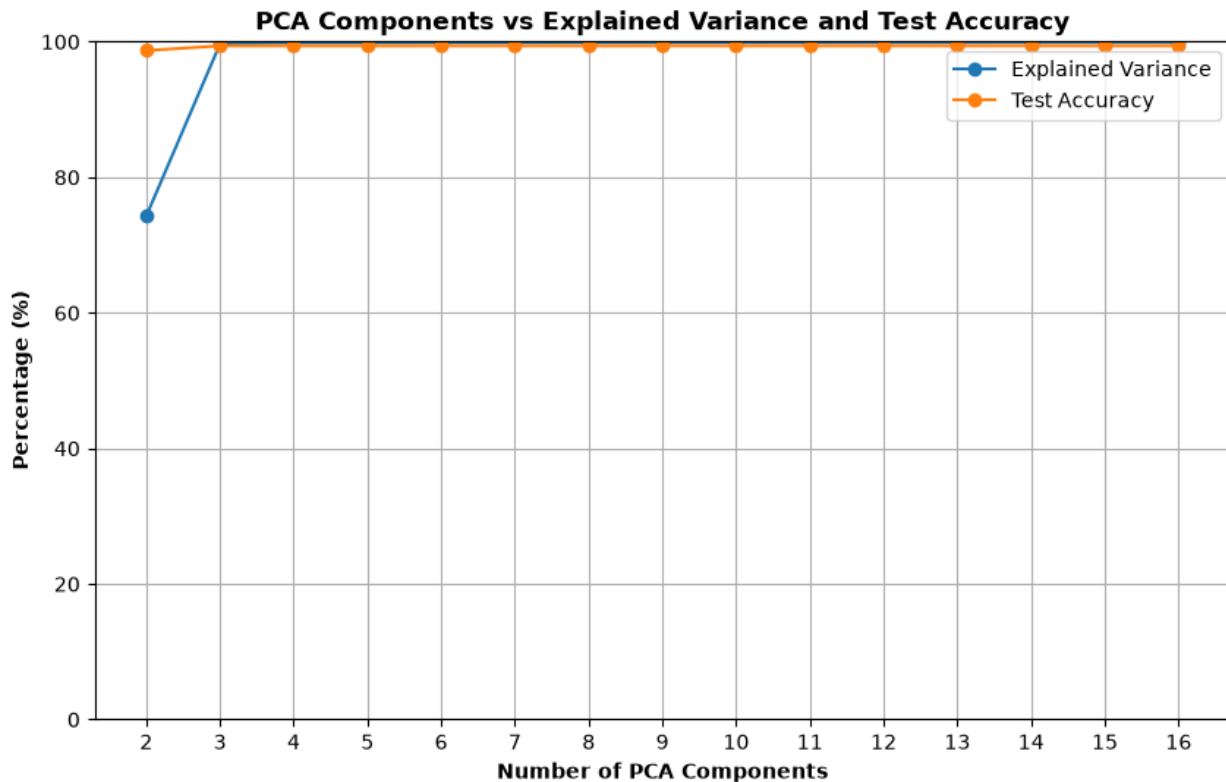
## Application Session:

- A. QSVC
- B. VQC

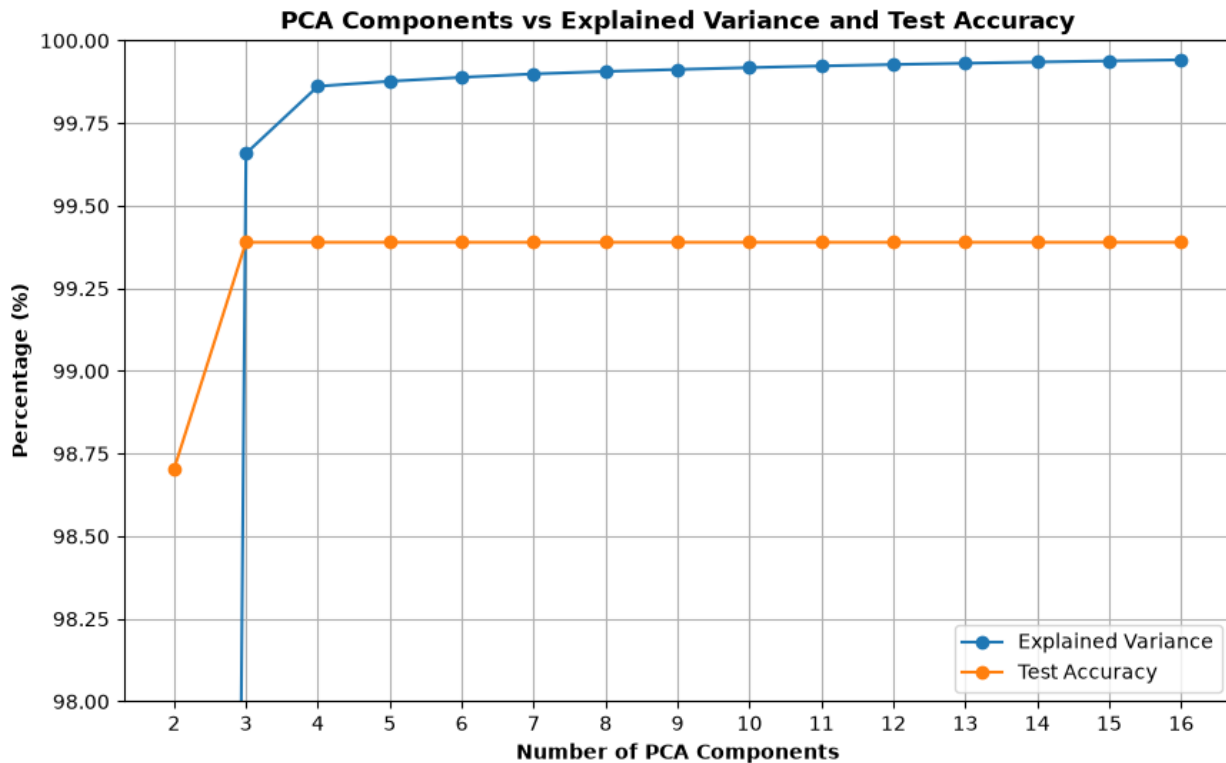


**See Jupyter Notebooks**

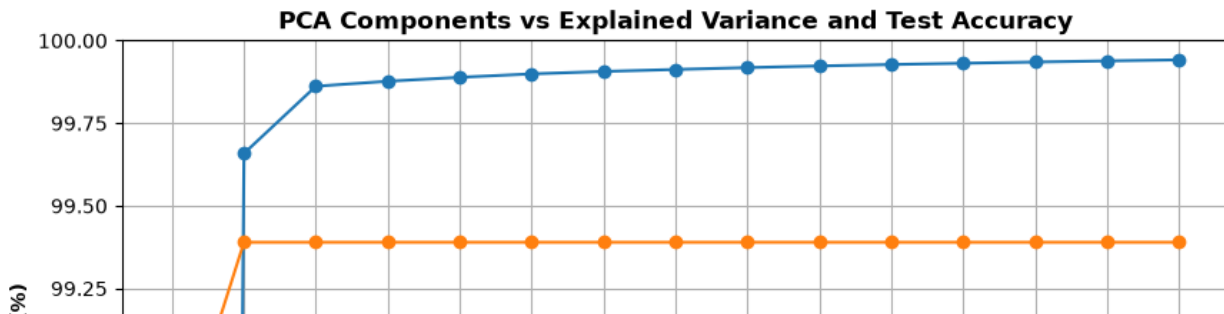
# Dimensionality Reduction: PCA & RBF-SVM



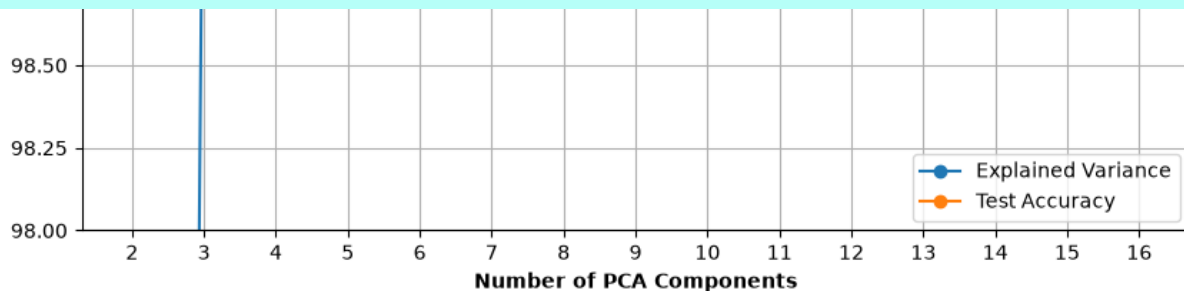
# Dimensionality Reduction: PCA & RBF-SVM



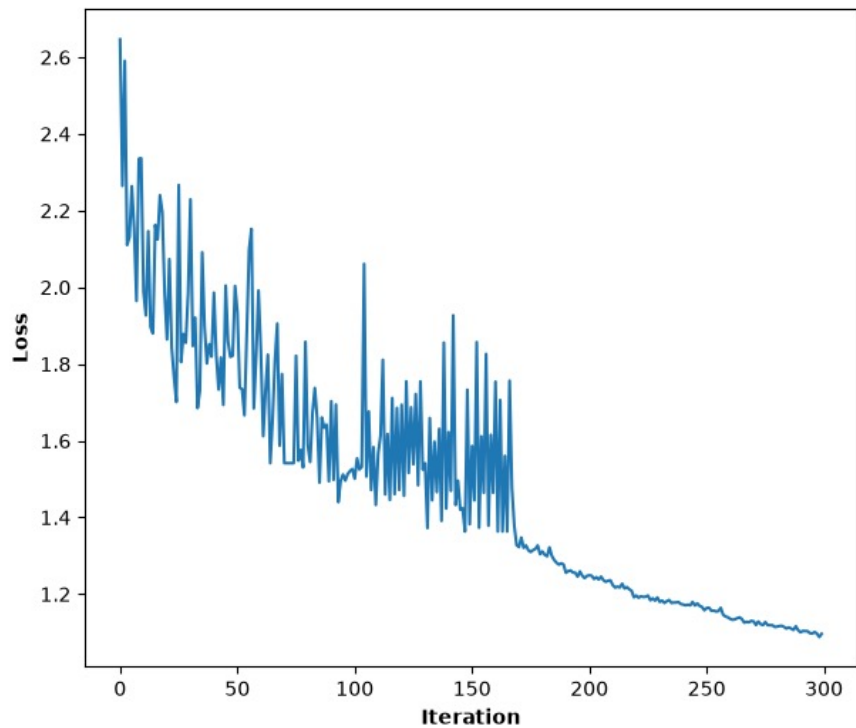
# Dimensionality Reduction: PCA & RBF-SVM



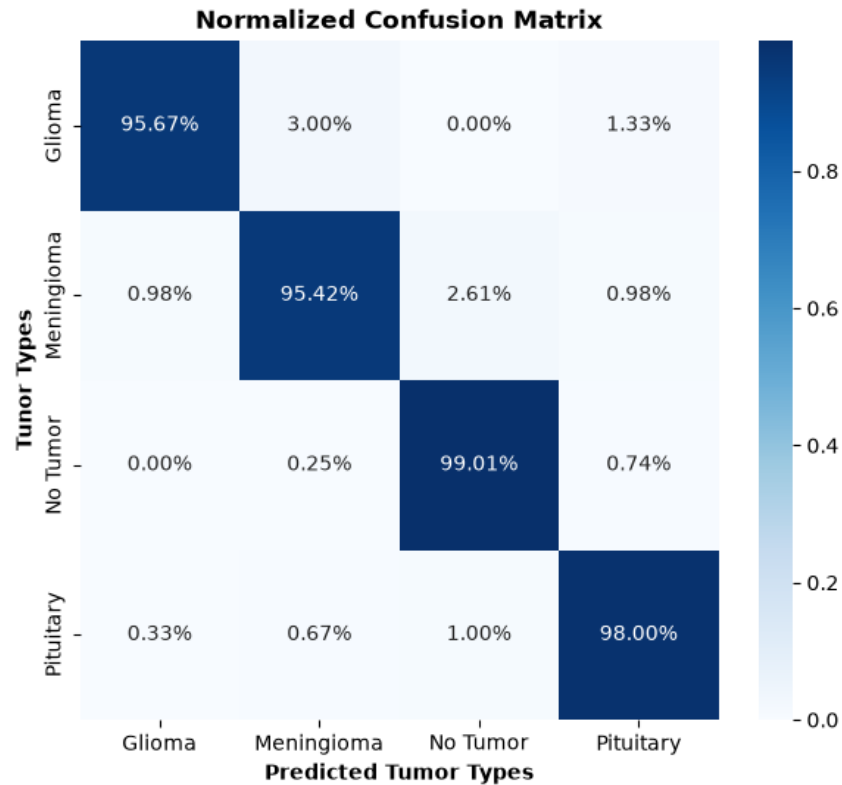
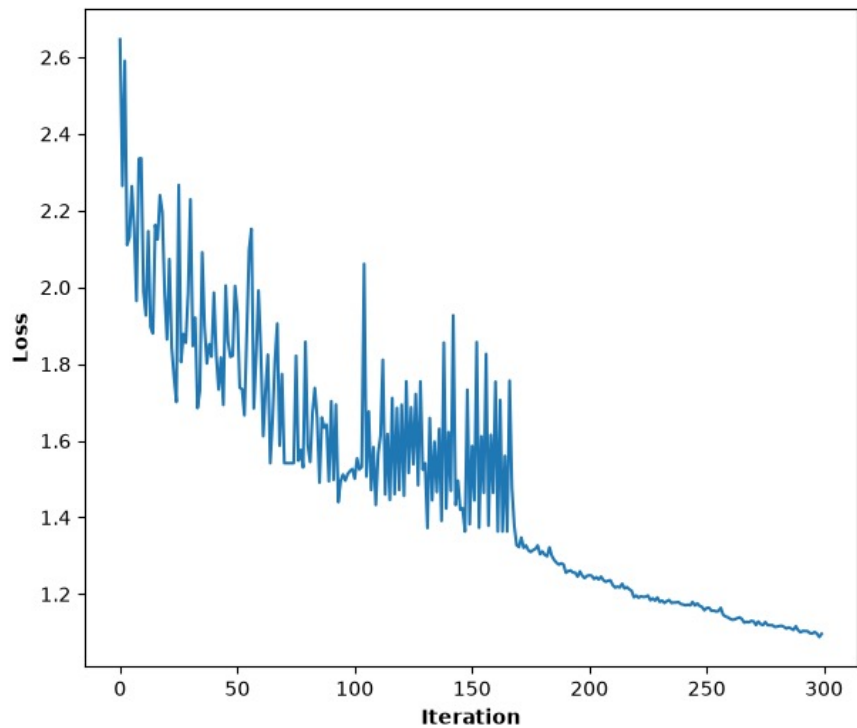
Most of the key feature information from the initial 128 dimensions is contained within the first 4 PCA components!



# VQC : Results



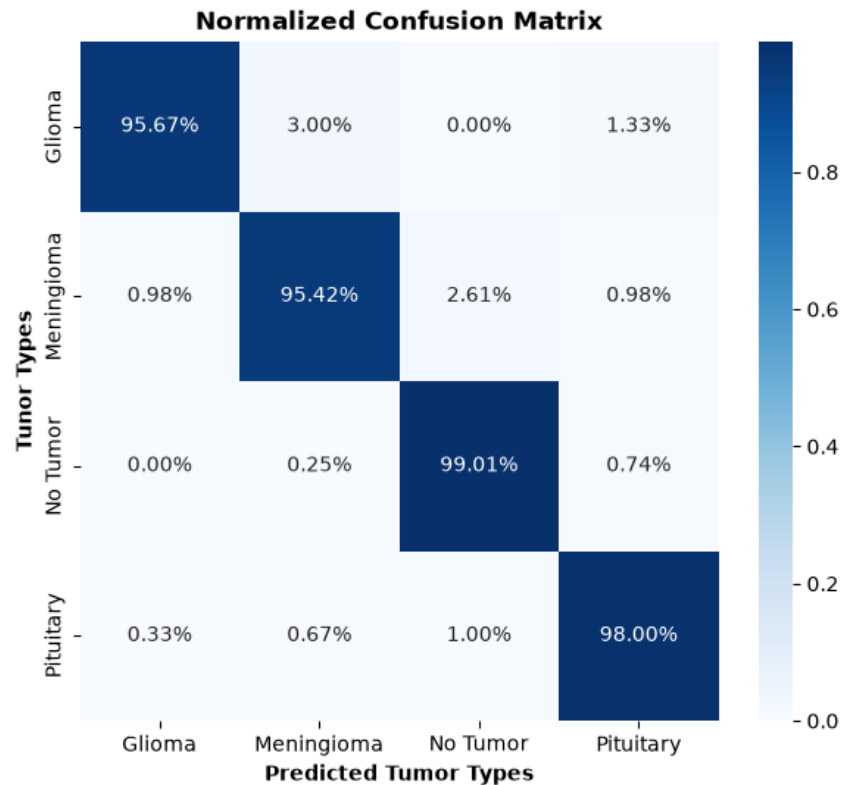
# VQC : Results



# VQC : Results

## Performance Metrics:

|                     | Precision   | Recall      | F1-Score     | Support     |
|---------------------|-------------|-------------|--------------|-------------|
| <b>Glioma</b>       | <b>0.99</b> | <b>0.96</b> | <b>0.97</b>  | <b>300</b>  |
| <b>Meningioma</b>   | <b>0.96</b> | <b>0.95</b> | <b>0.96</b>  | <b>306</b>  |
| <b>No Tumor</b>     | <b>0.97</b> | <b>0.99</b> | <b>0.98</b>  | <b>405</b>  |
| <b>Pituitary</b>    | <b>0.97</b> | <b>0.98</b> | <b>0.97</b>  | <b>300</b>  |
| <b>Accuracy</b>     |             |             | <b>97.97</b> | <b>1311</b> |
| <b>Macro avg</b>    | <b>0.97</b> | <b>0.97</b> | <b>0.97</b>  | <b>1311</b> |
| <b>Weighted avg</b> | <b>0.97</b> | <b>0.97</b> | <b>0.97</b>  | <b>1311</b> |



# Circuit Depth Dependent Performance

| Qubits | Iterations | Accuracy | Macro Avg F1 | Weighted Avg F1 | Best Loss |
|--------|------------|----------|--------------|-----------------|-----------|
| 5      | 1200       | 95.65%   | 0.95         | 0.96            | 1.0112    |
| 5      | 300        | 81.39%   | 0.80         | 0.81            | 1.4648    |
| 4      | 300        | 97.18%   | 0.97         | 0.97            | 1.0881    |
| 3      | 300        | 82.07%   | 0.80         | 0.81            | 1.2466    |
| 3      | 1000       | 82.53%   | 0.80         | 0.82            | 1.2447    |
| 2      | 150        | 54.00%   | 0.35         | 0.38            | 1.5947    |

# Takeaways

- ❖ **QSVC model is a simple quantum approach that stands as a reference for more elaborated quantum models**
- ❖ **Achievement of high-performance metrics using VQC for tumor classification**
- ❖ **Challenge: Building a quantum circuit that enables feature expressivity and trainability without being expensive computationally**

# Perspectives

- ❖ **Fine-tuning the current model to optimize the VQC performance**
- ❖ **Running the model on a real hardware Quantum Runtime**
- ❖ **Development of QCNN model**
- ❖ **Exploring Quantum Computing Approaches beyond Medical Applications**

# Acknowledgement : The NeuroVision Team

**Dr. Valentina Disarlo (Mathematics)**

**Ms. Harshita Mishra (Bioinformatics)**

**Dr. Julia Nowak (Neuroscience)**



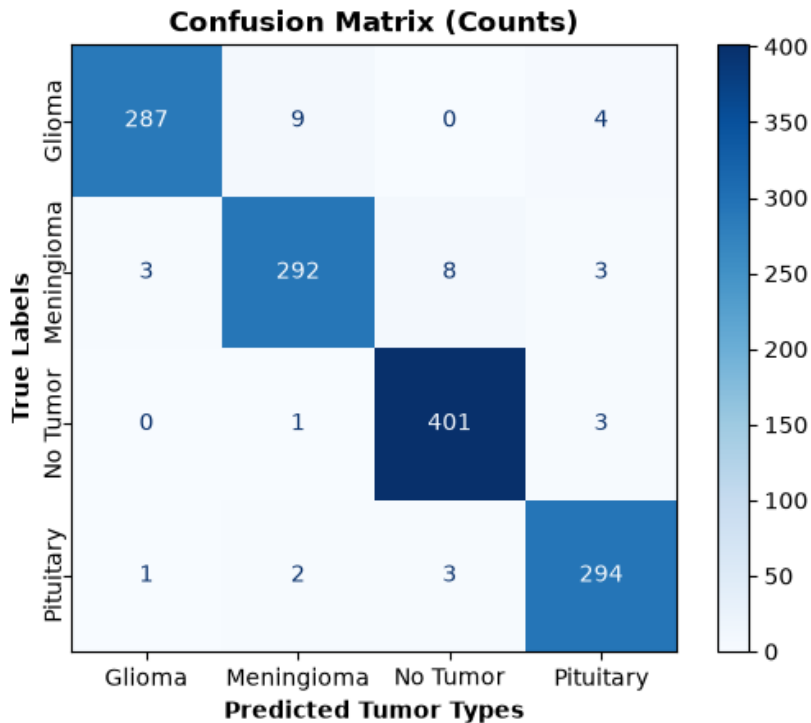
**Dr. Boubacar Tanda Bonkano (Physics)**

# Q&A Session

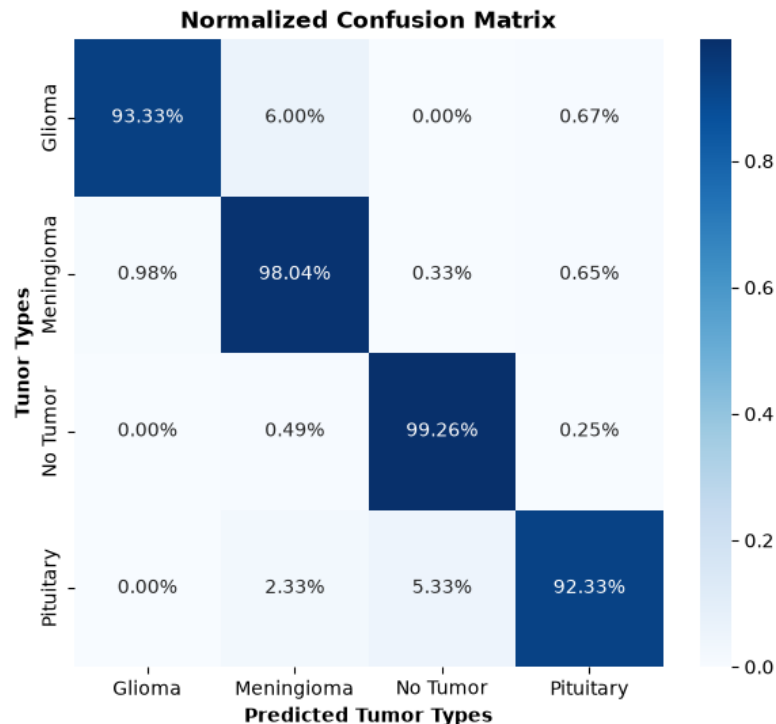
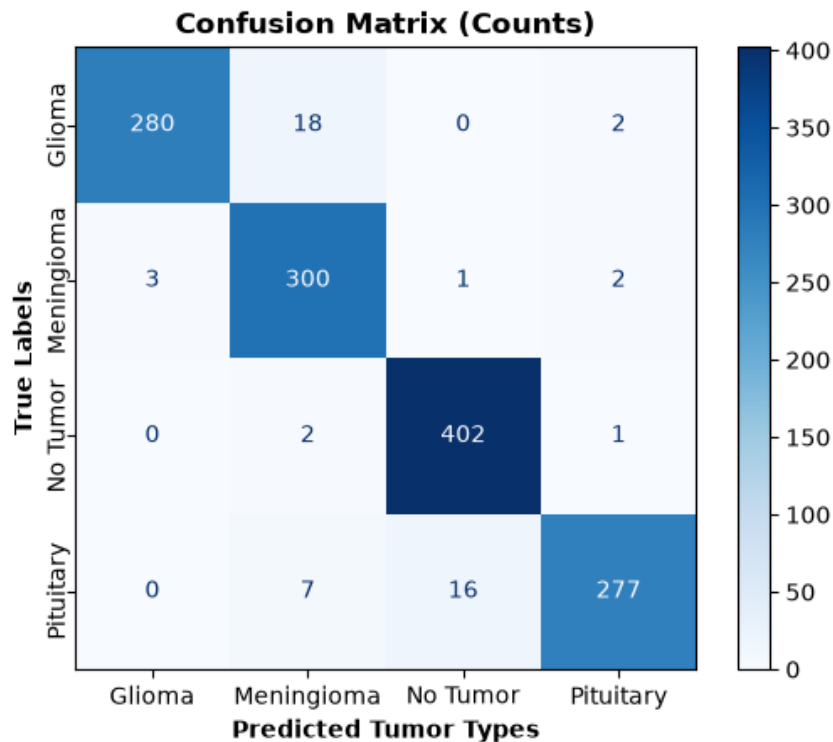
Thank you for your  
attention!



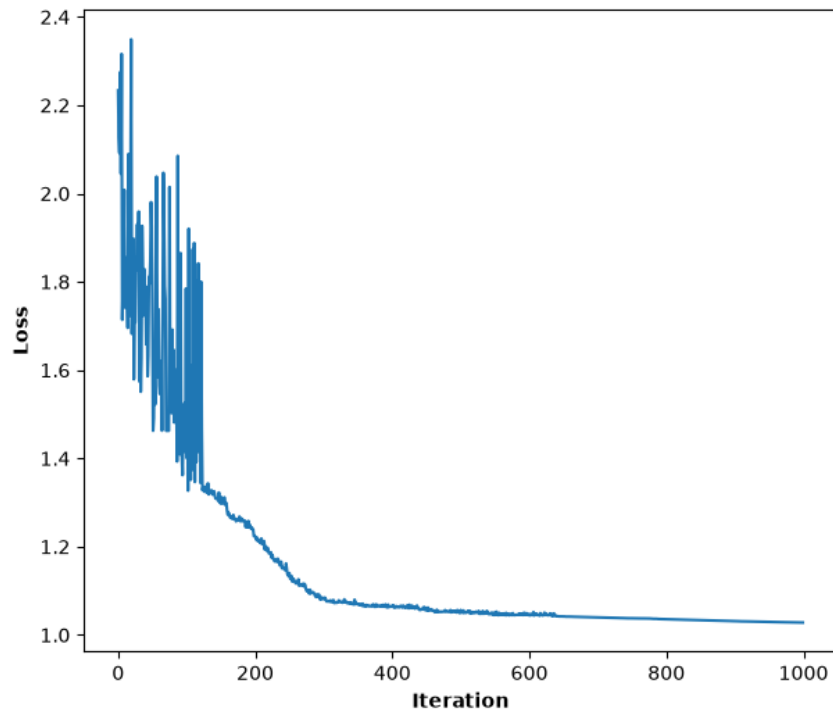
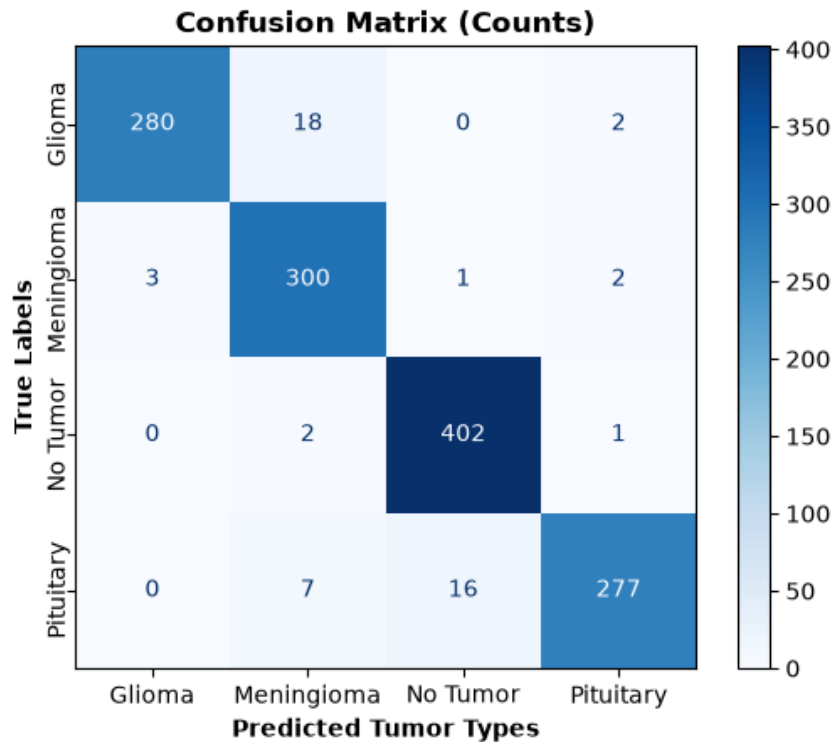
# VQC : Results (COBYLA Maxiter=300)



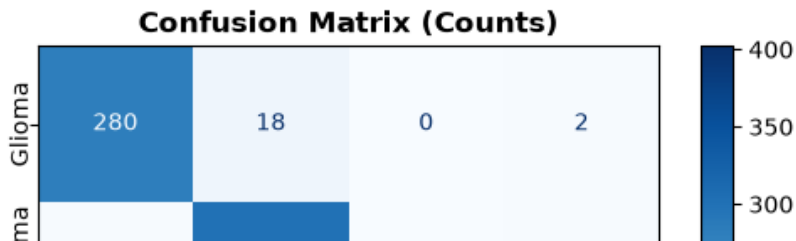
# VQC : Results (COBYLA Maxiter=1000)



# VQC : Results (COBYLA Maxiter=1000)

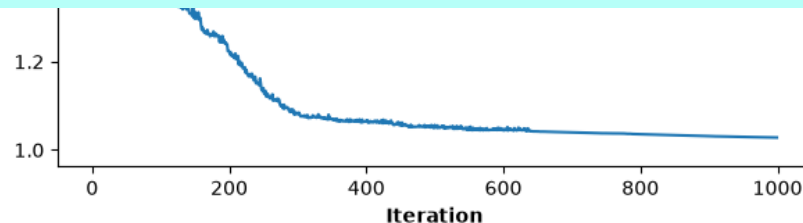
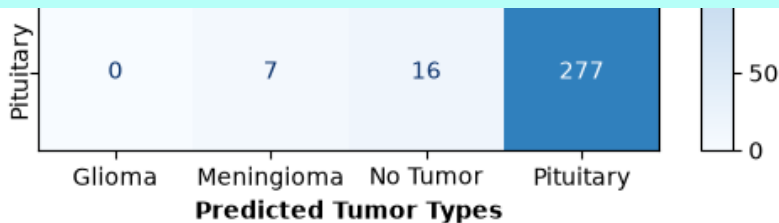


# VQC : Results (COBYLA Maxiter=1000)



**Beware of overfitting effects!**

**Longer training does not necessarily lead to better model performance.  
Implement Appropriate Checkpoints for a convenient Stopping.**



# Training Parameters And Qiskit Packages

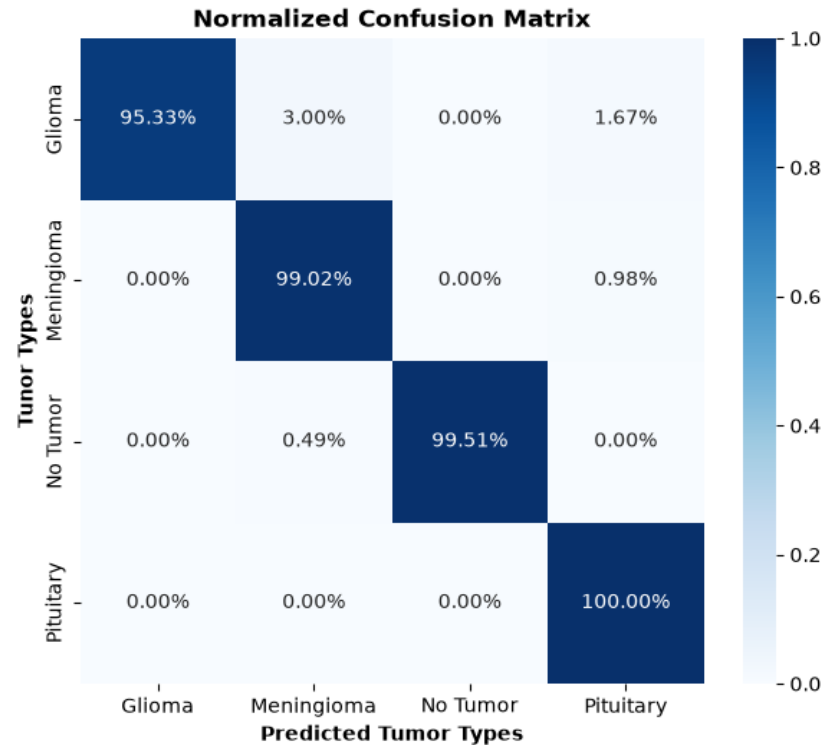
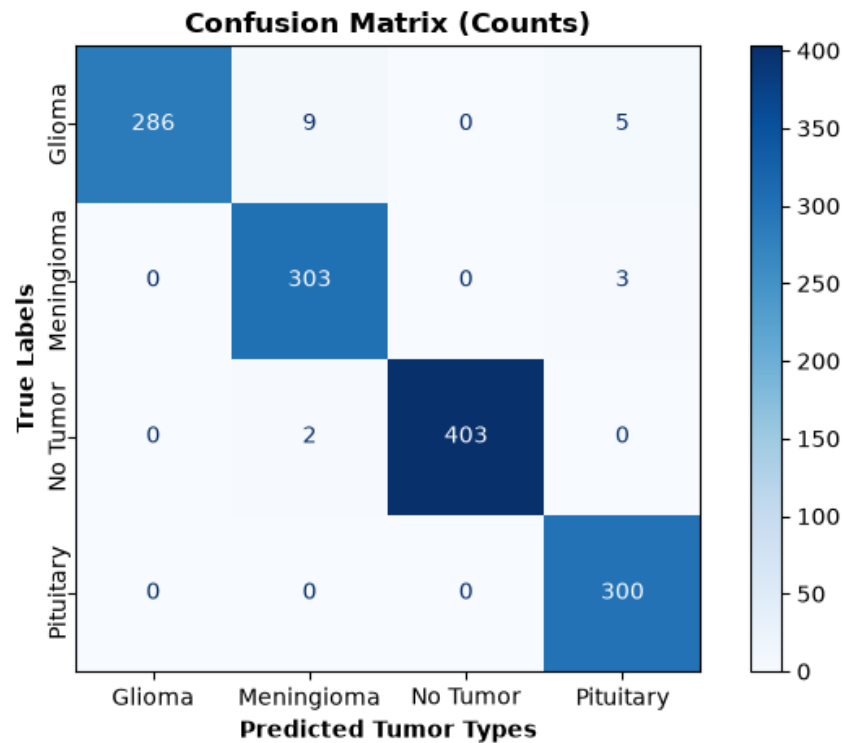
Training pool: (5712, 128)  
Test set: (1311, 128)  
Original training distribution: [1321 1339 1595 1457]  
Test distribution: [300 306 405 300]  
Selected training distribution: [1318 1336 1592 1454]  
Selected training samples: (5700, 128)

Feature Map : ZZFeatureMap  
Ansatz : ExcitationPreserving  
Optimizer : COBYLA  
Model : VQC

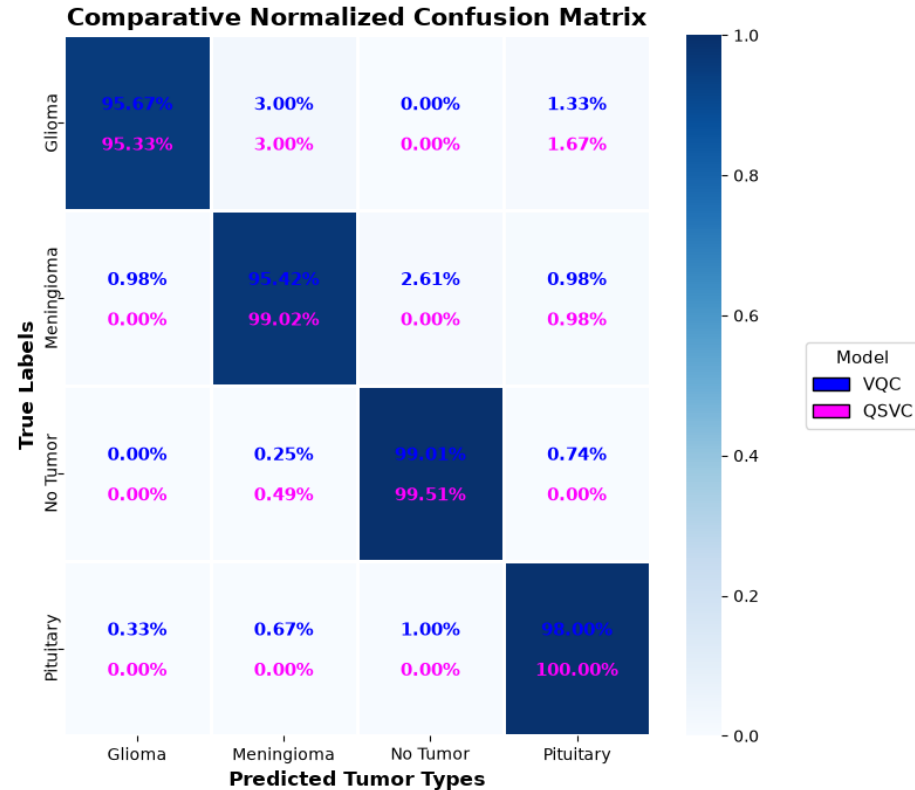
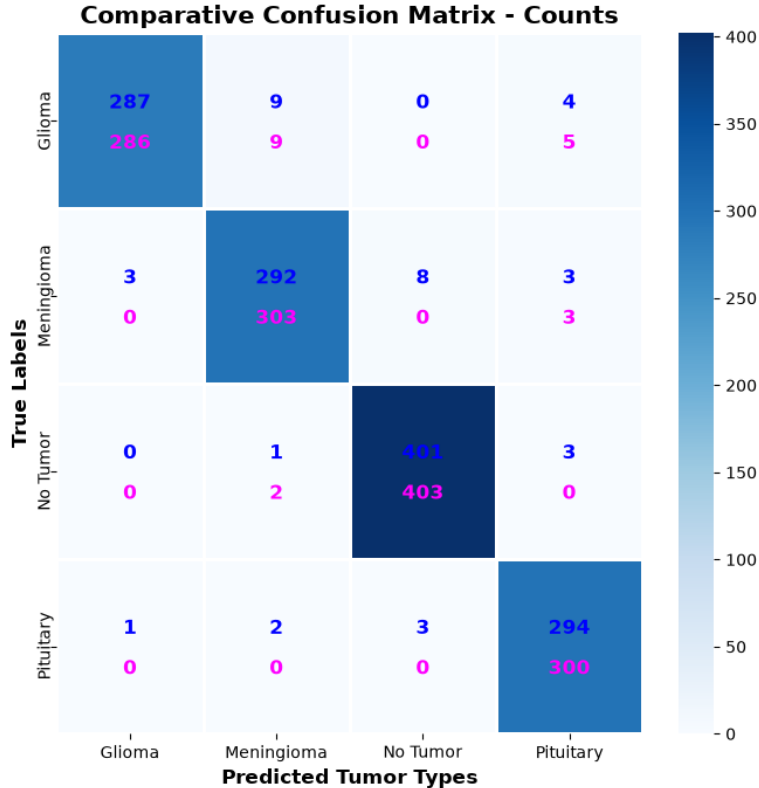
Qiskit: 1.2.4  
Qiskit ML: 0.7.2  
Qiskit Aer: 0.14.2  
Qiskit Algorithms: 0.3.0



# QSVC

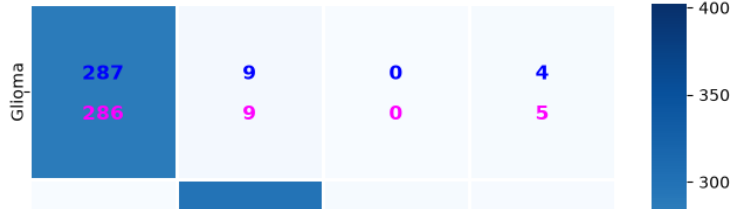


# VQC vs QSVC

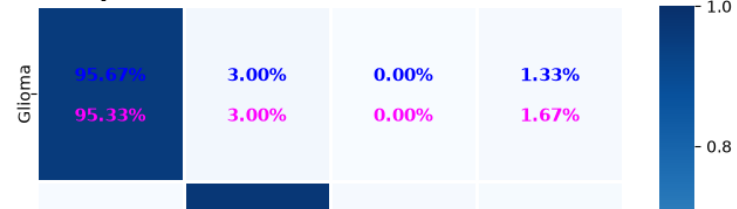


# VQC vs QSVC

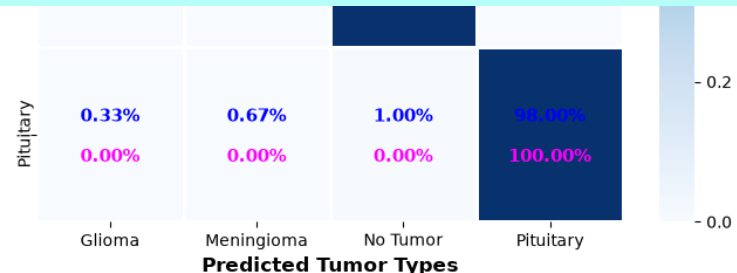
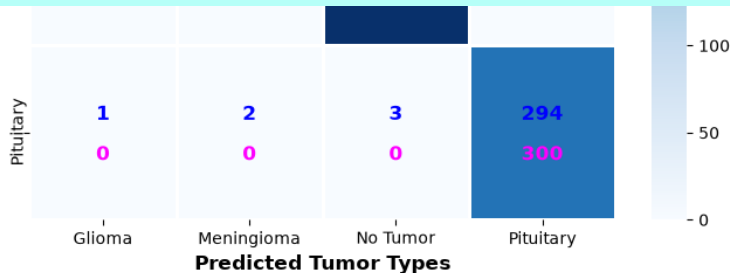
Comparative Confusion Matrix - Counts



Comparative Normalized Confusion Matrix



Despite the complexity residing in the feature mapping onto the quantum circuit, it is possible to achieve a good performance with the VQC model. A fine-tuning with COBYLA and different optimizers will help to match the QSVC performance which is within reach.



Qiskit Demo – presented on 21 August 2026

Terminus!